Chapter 63

Analog Voltage Comparator

63.1 Introduction

ESP32-P4 integrates two analog voltage comparators, i.e., analog voltage comparator *0* and analog voltage comparator *1*. These comparators rely on special pads that support voltage comparison functionality to monitor voltage changes on these pads. Each analog voltage comparator has two pads associated with it, for the main voltage and the reference voltage respectively. The voltage comparison result generated by the analog voltage comparator can be used as Event Task Matrix (ETM) events to drive ETM tasks of other peripherals or trigger interrupts.

Note:
The information provided in this chapter applies to both analog voltage comparator *0* and analog voltage comparator *1*. Unless otherwise indicated, the analog voltage comparator in this chapter refer to both analog voltage comparator *0* and analog voltage comparator *1*.

63.2 Feature List

The analog voltage comparator has the following features:

- Voltage comparison
  - Configurable voltage comparison mode
  - Configurable reference voltage
- Interrupt upon changes of voltage comparison result
- ETM event generation

63.3 Architectural Overview

Figure 63.3-1 shows the structure of the analog voltage comparator.

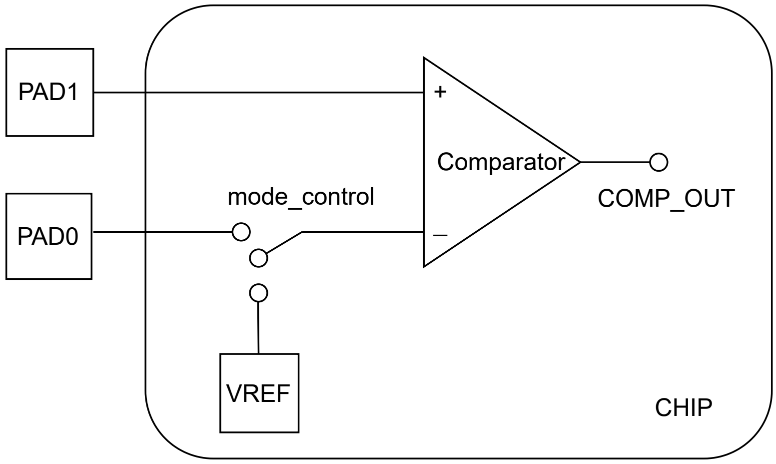

Figure 63.3-1. Analog Voltage Comparator Architecture

As shown in the figure above, the analog voltage comparator consists of:

- Two pad interfaces:
  - PAD1: Main voltage interface
  - PAD0: External reference voltage interface
- VREF: Internal reference voltage interface
- mode_control: Register-controlled switch for selecting the reference voltage source
- COMP_OUT: Comparator voltage comparison result, which is an on-chip internal signal that cannot be directly accessed externally
  - If the main voltage is higher than the reference voltage, the COMP_OUT signal is high.
  - If the main voltage is lower than the reference voltage, the COMP_OUT signal is low.

63.4 Functional Description

The analog voltage comparator has the following functions:

- **Voltage comparison**

  The analog voltage comparator relies on a specialized pads that support voltage comparison. For more information about general-purpose pads, please refer to Chapter 9 GPIO Matrix and IO MUX. Table below shows the mapping between the PAD of analog voltage comparator *0* and GPIO pins.

Table 63.4-1. Mapping Between PAD and GPIO

Analog Voltage Comparator	PAD0	PAD1
Analog voltage comparator *0*	GPIO51	GPIO52
Analog voltage comparator *1*	GPIO53	GPIO54

The comparison mode and reference voltage are configurable. For detailed configuration procedures, see Section 63.7 Programming Procedures.

The voltage comparison results are output as the COMP_OUT signal.

- **Voltage comparison mode control**

  The analog voltage comparator compares the main voltage with the reference voltage, which can either be internal or external. There are two voltage comparison modes depending on the reference voltage selection:

  – Comparing the main voltage with the external reference voltage.

  – Comparing the main voltage with the internal reference voltage.

  The voltage comparison mode can be configured through LPSYSREG_MODE_COMP*n* (*n* = 0 ~ 1).

- **Reference voltage configuration**

  The internal reference voltage range is 0 ~ (0.7 × VDD_IO_6) V with a step of 0.1 × VDD_IO_6 V, where VDD_IO_6 is the supply voltage of the analog voltage comparator which is equal to the power supply voltage of the chip (usually 3.3 V). The internal reference voltage value can be configured through LPSYSREG_DREF_COMP*n* (n*n* = 0 ~ 1).

  The external reference voltage is accessed directly from the pad with an input range of 0 ~ 0.7 x VDD_IO_6 V, same as the internal reference voltage, requiring no additional configuration.

- **Voltage comparison interrupt processing**

  The voltage comparison result, i.e., the COMP_OUT signal, can be either high or low. Whenever the output value changes, a corresponding interrupt signal is generated.

63.5 Event Task Matrix Feature

The analog voltage comparator on ESP32-P4 supports the Event Task Matrix (ETM) function, which allows analog voltage comparator ETM events to trigger any peripherals' ETM tasks. This section introduces the ETM events related to analog voltage comparator. For more information, please refer to Chapter 13 Event Task Matrix (ETM).

The analog voltage comparator can generate the following ETM events:

- GPIO_EVT_ZERO_DET_POS*n*: Indicates that the COMP_OUT signal changes from low level to high level, i.e., the main voltage changes from below to above the reference voltage.
- GPIO_EVT_ZERO_DET_NEG*n*: Indicates that the COMP_OUT signal changes from high level to low level, i.e., the main voltage changes from higher to lower than the reference voltage.

The analog voltage comparator does not support any ETM tasks.

63.6 Interrupts

ESP32-P4's analog voltage comparator can generate the GPIO_PAD_COMP_INT interrupt signal that will be sent to the Interrupt Matrix.

There are several internal interrupt sources from analog voltage comparator that can generate the above interrupt signal. The interrupt sources from analog voltage comparator are listed with their trigger conditions and the resulted interrupt signal(s) in Table 63.6-1.

Table 63.6-1. Analog Voltage Comparator's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
GPIO_COMP*n*_ALL_INT (*n* = 0 ~ 1)	Any COMP_OUT signal transition occurs	GPIO_PAD_COMP_INT
GPIO_COMP*n*_NEG_INT (*n* = 0 ~ 1)	The COMP_OUT signal changes from high level to low level	GPIO_PAD_COMP_INT
GPIO_COMP*n*_POS_INT (*n* = 0 ~ 1)	The COMP_OUT signal changes from low level to high level	GPIO_PAD_COMP_INT

Note:
For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter 12 *Interrupt Matrix* > Section 12.2 *Interrupt Terminology in ESP32-P4*.

Each interrupt source can be configured by a common set of registers that are described in Section *Interrupt Configuration Registers*. The specific registers can be found in Section 9.19.1 *HP GPIO Matrix Register Summary*.

63.7 Programming Procedures

The programming procedure for the analog voltage comparator is as follows:

Note:
For more information about the register fields mentioned in this section, refer to Chapter 20 System Registers (SYSREG).

1. Enable the comparator by setting LPSYSREG_XPD_COMP*n* (*n* = 0 ~ 1) to 1.
2. Disable normal digital pad functions (including IE, OE, WPU, and WPD) for PAD1, and if using the external reference voltage, also for PAD0. For detailed configuration, see Chapter 9 GPIO Matrix and IO MUX.
3. Configure the comparison mode by setting LPSYSREG_MODE_COMP*n* (*n* = 0 ~ 1):
   - 0: Configures to compare the main voltage with the internal reference voltage;
   - 1: Configures to compare the main voltage with the external reference voltage.
4. Configure the internal reference voltage through LPSYSREG_DREF_COMP*n* (*n*n = 0 ~ 1), which offers a voltage range of 0 ~ (0.7 × VDD_IO_6) V with a step of 0.1 × VDD_IO_6 V.